Drive-law noise reduction: fw v35 vs fw v37

Scale-car DC balancer board, Pi rosbag and SD-log analysis. Ricky Tan, 2026-11-06.

1. Summary

The fw v37 drive law lowers the drive-loop crossover from 2.76 Hz to 1.04 Hz. On the vehicle, it reduces the once-per-revolution (1x) flywheel line to 0.25–0.28 of its fw v35 level on the bus currents, the bus voltage and the FC share ratio. The design ratio is 0.29. Content above about 7 Hz is unchanged. A 1 kHz SD-card log shows that most of the remaining bus-voltage noise in the Pi logs is high-frequency content folded by the 21 ms telemetry snapshot. Its largest source is real bus current at the motor's 6x electrical frequency, which is power ripple from the motor's non-sinusoidal back-EMF. The RT1987 ideal diodes and the boost switching contribute little.

2. Data and method

Two runs of the same SDP drive cycle are compared: `rosbag_sdp_2026-10-30_01-49-38` (fw v35) and `rosbag_sdp_2026-11-05_20-09-52` (fw v37 rev 2). The share controller and all other run-path code are identical in the two builds. Statistics use only samples with both sources on the bus and $I_{tot} > 0.3$ A. The analysis excludes ± 0.5 s around each H-20 purge and $-0.2/+1.0$ s around each share-command step. Duplicate telemetry rows are removed, which gives a sample rate of 47.62 Hz. The share is $s = I_{fc}/I_{tot}$.

3. Results

Table 1 gives the 1x amplitude from a tracked-phase lock-in at 1.8–2.9 m/s.

1x amplitude (peak)	fw v35	fw v37	Ratio
I_{tot}	213 mA	54 mA	0.25
I_{batt}	187 mA	47 mA	0.25
I_{fc}	34 mA	9 mA	0.27
V_{bus}	14.2 mV	3.9 mV	0.28
Share ratio	0.032	0.008	0.25
v_{actual}	18.6 mm/s	12.1 mm/s	0.65

Table 1. 1x line at cruise (1160 / 1264 samples).

The bus-side 1x gain I_{tot}/v_{actual} fell to 0.33, against the design controller-gain ratio of 0.34, so the board runs the new law as designed. The v_{actual} 1x fell more than the new law alone predicts (0.65 against 0.85). The once-per-revolution disturbance was therefore smaller in the fw v37 run, for a reason the bags do not show.

The bus voltage follows $V_{bus} \approx V_0 - Z\, I_{tot}$, where Z is the bus impedance that the two droops set in parallel. Table 2 gives Z from the $I_{tot} \rightarrow V_{bus}$ cross-spectrum and the fraction of the V_{bus} variance that I_{tot} explains.

Run	Band	$\lvert Z \rvert$ (mΩ)	V_{bus} rms (mV)	Explained by I_{tot}
fw v35	2–6 Hz	57	10.8	67 %
fw v35	6–24 Hz	41–47	7.0–8.7	9–12 %
fw v37	2–6 Hz	41	5.4	21 %
fw v37	6–24 Hz	–	5.5–8.1	5–7 %
Battery only	2–6 Hz	159	23.1	92 %

Table 2. Bus impedance and the droop-driven part of V_{bus}, clean cruise windows.

At the 1x line, V_{bus}/I_{tot} is 67 mΩ before and 73 mΩ after. The V_{bus} 1x reduction is therefore the I_{tot} 1x reduction passed through an unchanged droop. This impedance is about 0.2 of the design droop of 0.30–0.36 Ω. In fw v37, most of the remaining V_{bus} noise in the bag is not coherent with I_{tot}. Section 4 shows that this noise is mostly high-frequency content folded by the telemetry sampling.

Figure 1 shows the cruise spectra. The 1x peak falls by about one decade in power on every channel, including the share ratio. Above about 7 Hz the two runs overlap.

The share ratio appeared not to improve because two terms that the drive law does not touch dominate it: the H-20 purges and the SDP command steps. Figure 2 shows the share 2–6 Hz rms as each term is removed. With both removed, the share noise falls from 0.021 to 0.010.

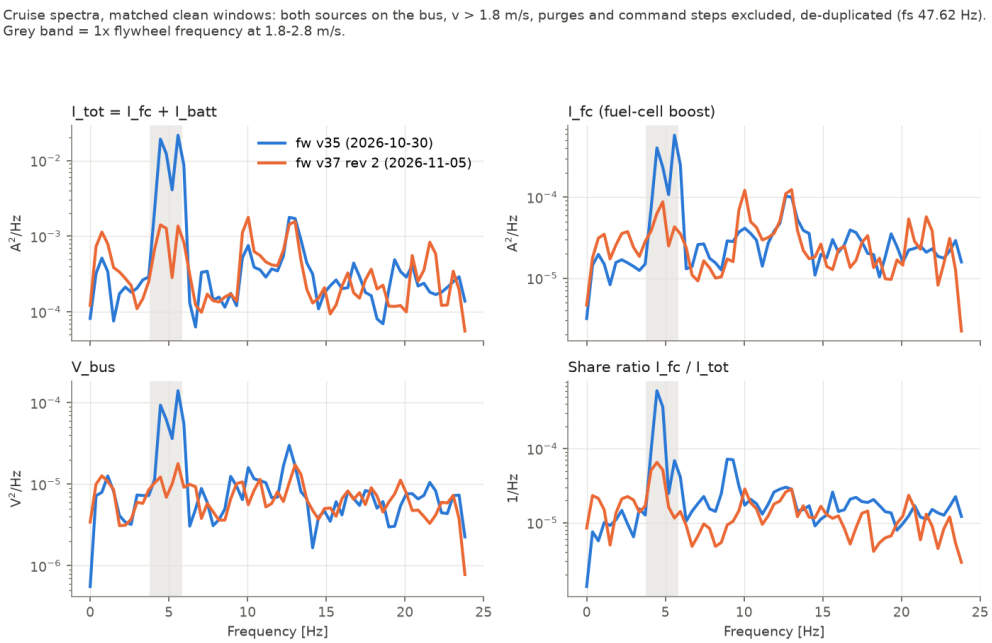


Figure 1. Cruise PSDs (v $>$ 1.8 m/s), fw v35 (blue) and fw v37 (orange). Grey band: 1x frequency at 1.8–2.8 m/s.

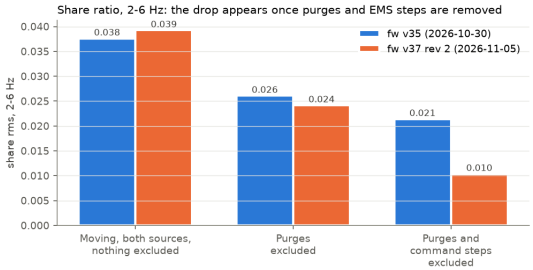


Figure 2. Share-ratio rms, 2–6 Hz, under successive masks.

Figure 3 breaks down the uncommanded share n = s − cmd in the clean windows. The stacked areas give the part of n that is coherent with the 1x rotation angle, then with I_{tot}, then with V_{fc}. In the 2–6 Hz band, the 1x-locked part falls from 0.018 to 0.007 rms. The governor floor reference, 0.125 A / I_{tot} filtered over 19.5 ms, follows n closely in this band (0.026 against 0.025 before, 0.011 against 0.011 after). The remaining 1x share term is therefore the setpoint clip at the minority-current floor, which the share loop tracks; it is not a loop deficiency. Above about 6 Hz both runs show a flat floor of about 0.011 rms, which Section 4 identifies as mostly folded content.

4. Remaining V_{bus} ripple: switching noise and the ideal diodes

The bag cannot resolve anything above its 23.8 Hz Nyquist frequency. This section therefore uses a 1 kHz SD-card log, ML0226 (fw v37, bench). Its FC input is a bench supply (11.1 V, no purges), its battery input is the pack, and it contains closed-loop speed holds at 1.0, 1.5 and 2.0 m/s. The firmware loop and the log run at about 875 Hz (median tick 1131 μs), not 1 kHz.

Figure 4 shows the V_{bus} spectrum at the loop rate and what a 21 ms telemetry snapshot makes of it.

Three results follow from Figure 4:

- **Most of the bag's V_{bus} noise is folded.** The snapshot reads 2.1–8.5 mV against 0.5–3.7 mV of true in-band content; 72–98 % of the snapshot variance comes from above 23.8 Hz. A 21 ms mean removes the fold almost completely.
- **The standstill floor is the ADC chain.** At standstill, V_{bus}, V_{rgn}, I_{fc} and I_{batt} all read 0.42–0.50 LSB, with cross-channel correlation of 0.06 or less.
- **The dominant moving line is the motor's 6x electrical order.** It lies at 397 Hz per m/s, which is 6 times the

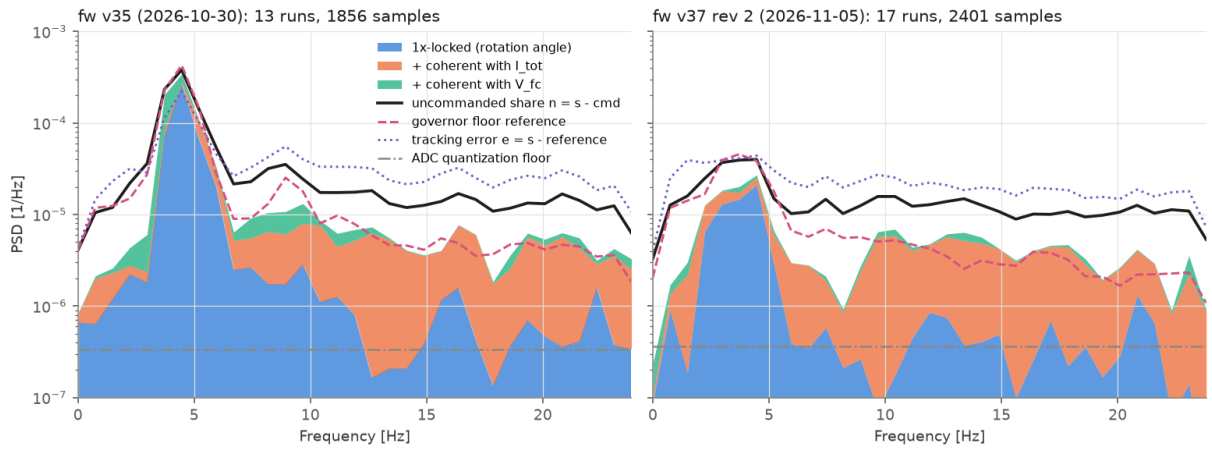


Figure 3. Uncommanded share PSD and its coherent parts, clean windows (both sources on the bus, v $>$ 0.3 m/s, purges and command steps excluded, de-duplicated).

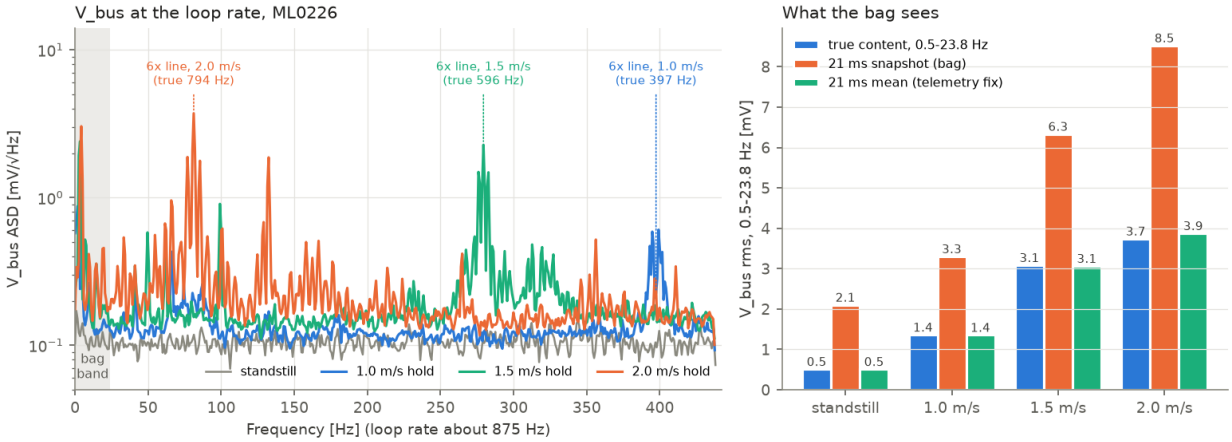


Figure 4. Left: V_{bus} amplitude spectral density in ML0226. The dotted markers show where the motor 6x electrical line lands after folding at the loop's Nyquist frequency. Right: V_{bus} rms in 0.5–23.8 Hz for the true signal, a 21 ms snapshot and a 21 ms mean.

electrical frequency of the 2-pole-pair motor, so it is real at 600–1000 Hz at cruise. It is coherent with I_{tot} at 0.94–0.99 and with the commanded motor current at only 0.01–0.05, so it is real current that the VESC draws from the bus. The bus impedance at this line is 81–133 mΩ, about 1.5 times its low-frequency value.

Figure 5 shows the whole log as a spectrogram. The dashed overlay is 397 Hz per m/s times the measured speed, folded at the 438 Hz Nyquist frequency. It follows the dominant line through every hold and every ramp, which identifies the line as the 6x electrical order. The fainter rising traces are lower motor orders.

Table 3 gives the resulting budget for the bag at fw v37 cruise (about 12.6 mV rms per sample). The split is measured on the bench and scaled to the vehicle; that scaling is not verified.

Source	mV rms	Confidence
Folded motor 6x electrical line (back-EMF power ripple, Section 5) and other orders	8–10	Medium
Folded broadband excess (likely VESC 15 kHz PWM alias)	5–6	Low–medium
ADC floor, folded	2.7–3.3	High
Real 1x droop response to I_{tot} (2–6 Hz)	2.4	High
TPS61288 switching and PFM ripple, aliased	$\leq$ 1.2	Medium
RT1987 ideal diodes	$\leq$ 0.6	Medium–high

Table 3. Budget of the V_{bus} noise in the bag band, fw v37 cruise. The quadrature sum of the central values is 12.3 mV.

The RT1987 ideal diodes add no measurable V_{bus} noise. Below about 1.75 A, which covers every rig current, each switch servos a forward drop of about 35 mV (datasheet p.7, Section 17.6), plus about 10 mΩ incremental. The

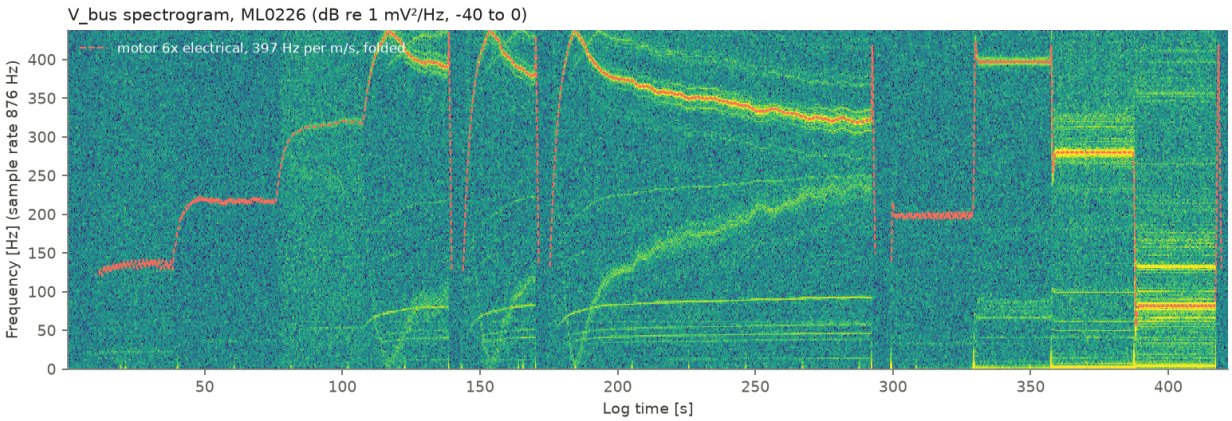


Figure 5. V_{bus} spectrogram of ML0226 (422 s, sample rate 876 Hz). Dashed: the motor 6x electrical line predicted from the measured speed, folded at the Nyquist frequency.

common 35 mV cancels in the share. No FC dropout or chatter occurs at the minority floor: every zero I_{fc} in the bags falls inside an H-20 purge. The servo bandwidth is not specified; above it each switch acts as about 35 mV / I, which is a candidate for the impedance rise at 600–800 Hz. This is not measured.

Boost switching contributes little to V_{bus}. The FC boost always runs in light-load (PFM) mode and the battery boost runs near its boundary (about 0.45 A). The ripple appears on I_{fc} and V_{fc}, upstream of the ideal diode, but only weakly on V_{bus}.

Consequence for filtering. A 20 ms mean in the telemetry would remove most of the folded noise in the bags. However, the loop itself samples at 875–1000 Hz, so the 6x line folds near DC in the control samples at about 2.2–2.5 m/s, the cruise speed. No digital filter after sampling can remove that alias. Only an analog filter at the ADC pin, or averaging over the full loop tick, prevents it.

5. Origin of the 6x electrical line

The 6x line is a deterministic power ripple locked to rotor angle, not noise. The VESC drives sinusoidal FOC currents, so the power it draws ripples at 6 times the electrical frequency only if a 5th or 7th harmonic distorts the product of back-EMF and current. Three mechanisms can do this, and each predicts a different scaling of the bus-current ripple amplitude A_6 with speed v and motor current I:

- non-sinusoidal back-EMF: $A_6 \propto v \cdot I$, a fixed fraction of the mechanical power;
- dead-time distortion in the VESC: $A_6 \propto I$, independent of speed;
- cogging, through the speed ripple it causes: $A_6 \propto I / v$.

A_6 was fitted in 2 s windows of ML0226 (294 windows, 0.3–2.0 m/s). The fit uses the logged timestamps, because the line is aliased and the tick interval is irregular. It tracks the line's phase from the smoothed speed, and it includes ± 12 rotation sidebands, because the real once-per-revolution speed ripple frequency-modulates the line. Figure 6 compares the result with the three predictions, each computed from the window's own speed and current.

The data support the back-EMF mechanism:

- **Frequency.** The freely fitted line frequency is 397.2 Hz per m/s, which equals 6 $\times$ 2 pole pairs $\times$ 15.84 gear ratio $\times$ 2.089 Hz per m/s = 397.1 Hz per m/s.
- **Speed at fixed current.** In the open-loop holds at 4.2 A (133 windows, 1.10–1.40 m/s), $A_6 \propto v^{0.99 \pm 0.10}$. Back-EMF harmonics predict v^1, dead time v^0 and cogging v^{-1}.
- **Ripple per ampere.** A_6/I rises eightfold over the speed range, from 1.5 to 11.8 mA/A. Dead time predicts a constant value.
- **Fraction of mechanical power.** Above about 1.1 m/s, k_6 is constant at 0.12–0.13, which is the back-EMF signature. At cruise the ripple is therefore about ± 12.5 % of the mechanical power, or 55 mA peak on the bus at 2.0 m/s.

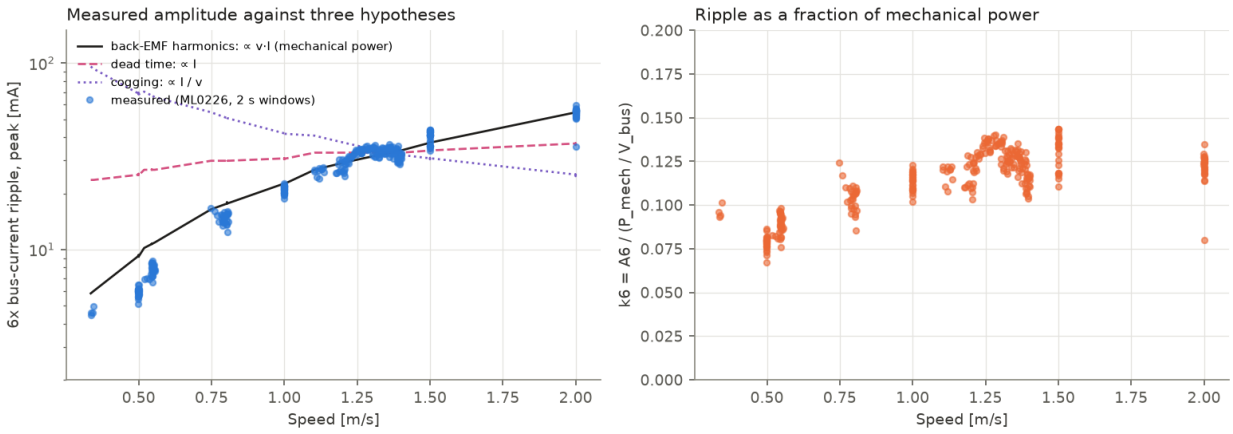


Figure 6. Left: 6x bus-current ripple amplitude in ML0226 against the three predictions, each normalised at the 1.3–1.45 m/s hold. Right: $k_6 = A_6 / (P_{mech} / V_{bus})$, with $P_{mech} = K_F \cdot I \cdot v$ and $K_F = 0.754$ N/A.

Three points are not settled. The current dependence on its own is not resolved, because speed and current move together in this log (I = 3.0–4.7 A). Below 1 m/s, k_6 is about 30 % lower, for an unidentified reason. In addition, a scope on the motor's line-to-line voltage while it coasts unpowered would measure the 5th and 7th back-EMF harmonics directly.

6. Conclusions

- The lower crossover reduced the 1x line to 0.25–0.28 on all bus channels and the share ratio, in agreement with the design.
- The share-ratio reduction is real. Purges and command steps hide it in unmasked data. The remaining 1x share term is the governor's minority-current floor, not a loop deficiency.
- V_{bus} at 1x is the droop image of I_{tot} through about 70 mΩ. The rest of the V_{bus} noise in the bags is mostly folded high-frequency content, largest at the motor's 6x electrical frequency. The ideal diodes and the boost switching contribute little.
- The 6x line is power ripple from non-sinusoidal motor back-EMF, about $\pm$12.5 % of the mechanical power at cruise. Dead time and cogging are ruled out as its main source.
- Content above 7 Hz did not change with fw v37. A 20 ms telemetry mean removes most of it from the bags. An analog filter at the ADC pins is needed to keep the 6x line out of the control samples.
- The tracking cost is visible: the speed-error rms below 0.7 Hz increased from 5.0 to 9.8 mm/s.
- A vehicle 1 kHz SD-card log with the H-20 and the pack is needed to confirm the budget at the vehicle operating point.